Example 8.21 From a well shuffled pack of 52 cards, one card is drawn at random. Find the probability of getting (i) red card (ii) heart card (iii) red king (iv) face card (v) number card

Suits of playing cards	Spade ♠	Heart ♥	Clavor ♣	Diamond ♦
Cards of each suit	A	A	A	A
	2	2	2	2
	3	3	3	3
	4	4	4	4
	5	5	5	5
	6	6	6	6
	7	7	7	7
	8	8	8	8
	9	9	9	9
	10	10	10	10
	J	J	J	J
	Q	Q	Q	Q
	K	K	K	K
Set of playing cards in each suit	**13**	**13**	**13**	**13**

Fig 8.5

Solution $n(S) = 52$

(i) Let A be the event of getting a red card.

$$n(A) = 26$$

Probability of getting a red card is

$$P(A) = \frac{26}{52} = \frac{1}{2}$$

(ii) Let B be the event of getting a heart card.

$$n(B) = 13$$

Probability of getting a heart card is

$$P(B) = \frac{n(B)}{n(S)} = \frac{13}{52} = \frac{1}{4}$$

(iii) Let C be the event of getting a red king card. A red king card can be either a diamond king or a heart king.

$$n(C) = 2$$

Probability of getting a red king card is

$$P(C) = \frac{n(C)}{n(S)} = \frac{2}{52} = \frac{1}{26}$$

(iv) Let D be the event of getting a face card. The face cards are Jack (J), Queen (Q), and King (K).

$$n(D) = 4 \times 3 = 12$$

Probability of getting a face card is

$$P(D) = \frac{n(D)}{n(S)} = \frac{12}{52} = \frac{3}{13}$$

(v) Let E be the event of getting a number card. The number cards are 2, 3, 4, 5, 6, 7, 8, 9 and 10.

$$n(E) = 4 \times 9 = 36$$

Probability of getting a number card is

$$P(E) = \frac{n(E)}{n(S)} = \frac{36}{52} = \frac{9}{13}$$

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